

Topic 1. Distribution laws of random variables

1.1. Distribution of random discrete variables

Example 1.1

Let us have a collection of 8 groups of 3 balls.

RRR	RRB	RBR	BRR
RBB	BRB	BBR	BBB

X number of red balls in the group.

$P(X)$ is a probability of finding a group with the number of red balls equal to X

Construct a distribution for a random variable X.

Example 1.2

Let us have a collection of 8 groups of 3 balls.

RRR	RRB	RBR	BRR
RBB	BRB	BBR	BBB

Y is number of blue balls in the group.

Construct a distribution for a random variable Y.

1.2 Checking the discrete probability distribution

Example 1.3

The farm has 15% white cows,

36% dark,

47% brown.

Is this probability distribution correct?

Example 1.4

The basket contains 33% red apples, 18% green, 62% yellow.

Is this probability distribution correct?

1.3 Calculation of the probability of a discrete random variable

Example 1.5

Let an incomplete probability distribution of a random variable be given

X	1	2	3	4	5
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P(X)	0.21	0.13	0.24	0.42	?
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Find $P(X=5)$

Find the probability of X greater than two.

$$P(X>2) = P(3) + P(4) + P(5) = 0.24 + 0.32 + 0.10 = 0.66$$

Example 1.6

Let an incomplete probability distribution of a random variable be given

X	0	1	2	3	4
P(X)	0.05	0.31	0.15	?	0.27

Find $P(X=3)$

Find the probability of X greater than two.

1.4 Mean (expected value) of a discrete random variable

Example 1.7

Let an probability distribution of a random variable be given

X	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

Find mean (expected value $E(X) = \mu_X$)

Example 1.8

Let an probability distribution of a random variable be given

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	0.10

Find mean or expected value $E(X) = \mu_X$)

1.5 Dispersion or variance and standard deviation of a discrete random variable

Example 1.9

Consider the probability distribution of a random discrete variable from Example 1.7

X	0	1	2	3
P(X)	0.125	0.375	0.375	0.125

Find variance (dispersion) and standard deviation.

Example 1.10

Consider the probability distribution of a random discrete variable from Example 1.2

Find mean or expected value $E(X) = \mu_X$.

Find variance (dispersion) and standard deviation.

Example 1.11

Consider the probability distribution of a random discrete variable from Example 1.8.

X	1	2	3	4	5
P(X)	0.21	0.13	0.24	0.42	0.10

Find variance (dispersion) and standard deviation.